Look Where It Matters: Distilling Vision Through Explanations

Can explainability provide a useful training signal
for distilling vision-language models?

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MOTIVATION

Can explanations become a training signal?

Knowledge distillation usually treats the teacher signal as a target to be matched **uniformly**

What if we **weight** the teacher signal by estimated importance?

Grad-CAM saliency tells us **which visual tokens** the teacher relied on most.

THE APPROACH

From uniform matching to saliency-weighted distillation

STANDARD KD

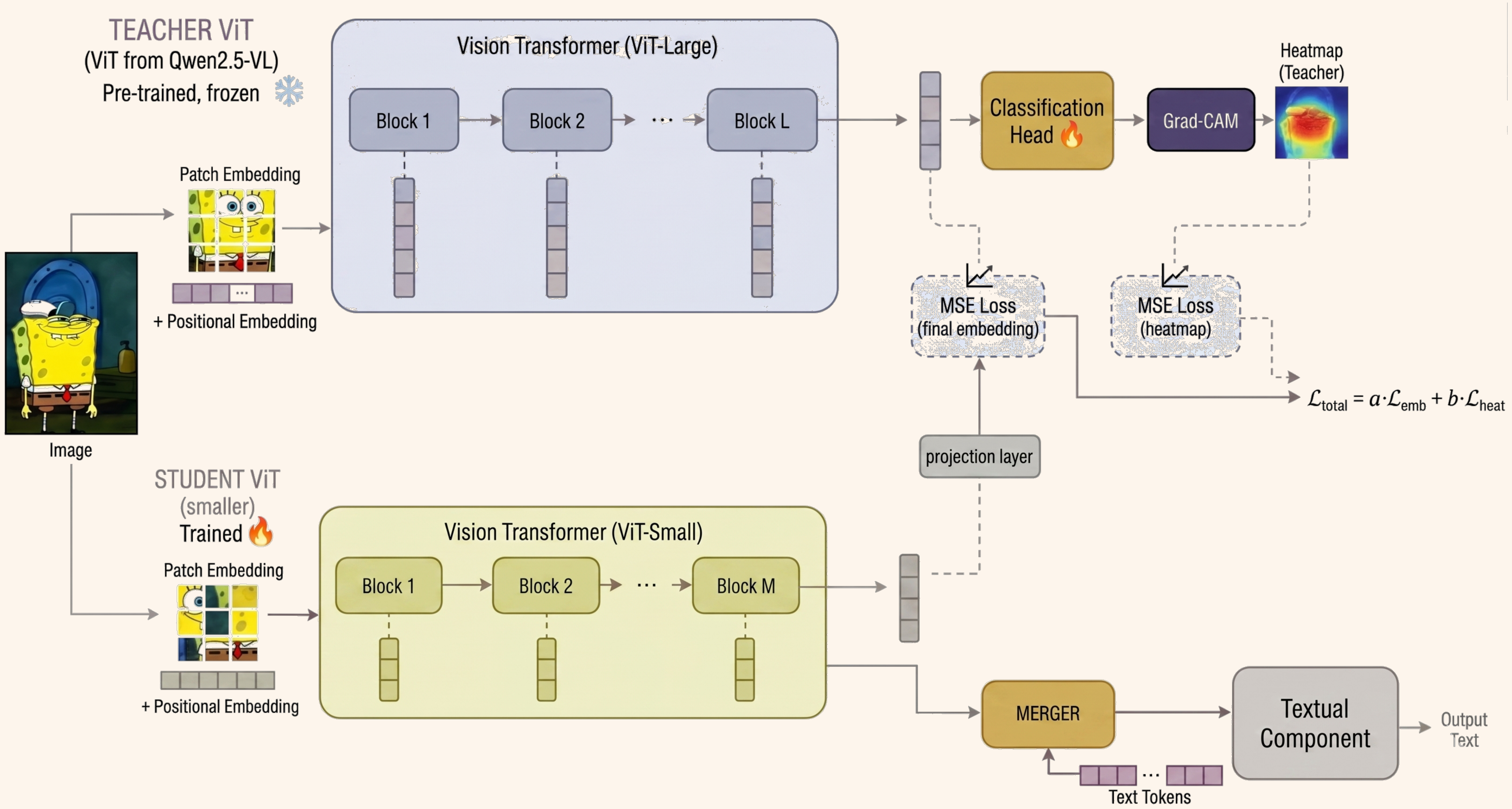
Match every teacher feature equally, with no notion of which features actually drive the teacher's decisions.

→

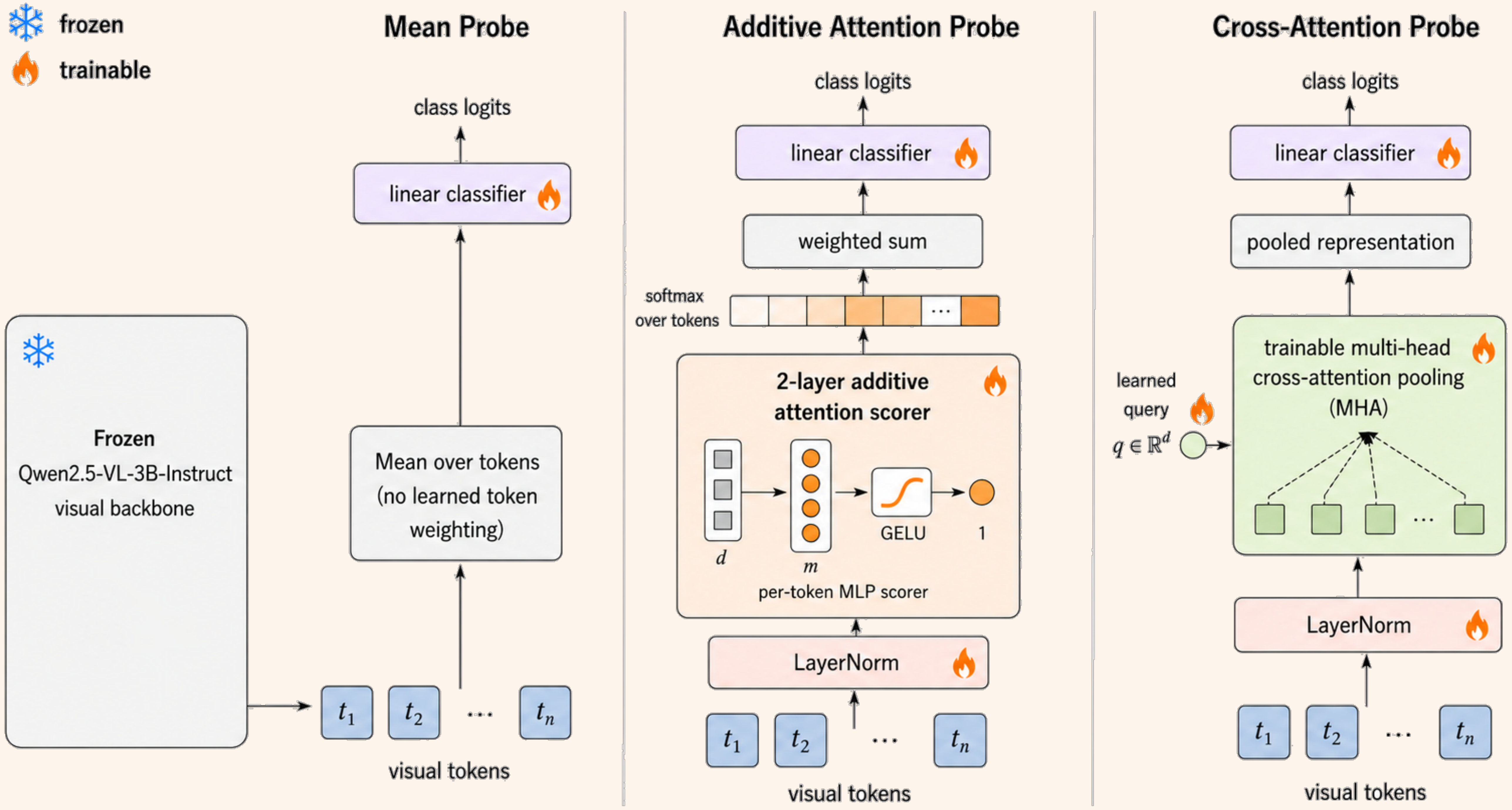
OUR APPROACH

Use Grad-CAM to weight tokens; the student focuses on preserving what the teacher considers **decision-relevant**.

Pipeline overview



Teacher probe architectures



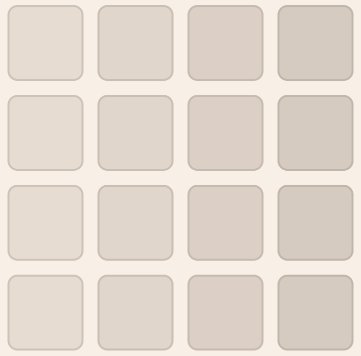
METHOD

From explanation to token weights

01 / FROZEN

Visual tokens

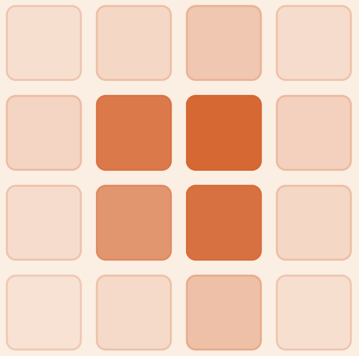
Qwen map: 28 x 28



02 / ATTRIBUTE

Probe saliency

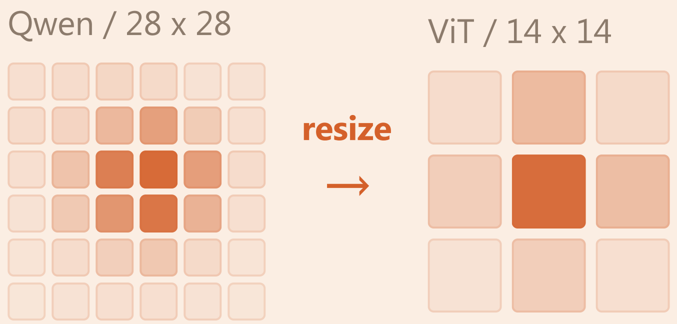
Class-relevant regions



03 / ALIGN

Map alignment

Saliency follows the new grid



METHOD

Three distillation objectives

BASELINE

Global MSE

$$\mathcal{L}_{\mathrm{global}} = \left\| \frac{1}{N} \sum_{i=1}^{N} s_i - \frac{1}{N} \sum_{i=1}^{N} t_i \right\|_2^2$$

SALIENCY-GUIDED

Explanation-weighted

$$\mathcal{L}_{\mathrm{expl}} = \sum_{i=1}^{N} A_i \| s_i - t_i \|_2^2$$

GLOBAL + LOCAL

Combined

$$\mathcal{L} = \lambda_g \, \mathcal{L}_{\mathrm{global}} + \lambda_e \, \mathcal{L}_{\mathrm{expl}}$$

Teacher probe accuracy

Probe type	Best epoch	Val accuracy	Test accuracy
Mean	10	**95.07%**	**95.00%**
Attention	2	94.93%	94.48%
Cross-attn	10	94.58%	94.78%

RESULTS

LLM jury

Variant	Student	Teacher	All agree	κ
Base MSE	10	427	**0.936**	0.312
Mean, expl-only	**69**	**367**	0.705	0.355
Attn, expl-only	56	381	0.741	0.367
Cross, expl-only	20	416	0.856	0.337
Mean, MSE+expl	58	379	0.737	0.355
Attn, MSE+expl	61	376	0.737	0.338
Cross, MSE+expl	37	400	0.817	**0.372**

Saliency retains visible texture

GLOBAL MSE

"The cap of the mushroom is light brown and has a slightly textured surface, with some areas appearing more raised or bumpy than others."

EXPLANATION-GUIDED

"The cap has a rough, textured surface with small, raised spots and ridges. The edges of the cap appear to be slightly curled inward."

READING

Guidance retains the reddish, raised cap texture that the baseline describes more generically.

Guidance recovers object identity

GLOBAL MSE

"The image shows a pair of red high-heeled shoes with white trim and a pointed toe."

EXPLANATION-GUIDED

"The image shows three bowls placed on a surface. The bowls are of different colors: one is red, another is brown, and the third is a combination of both colors."

READING

Guidance recovers the visible containers instead of hallucinating shoes.

Full scene understanding limitations

TEACHER VLM

"The image shows a person standing next to a large, detailed dinosaur skull mounted on a stand. The skull appears to be that of a Triceratops, characterized by its three prominent horns and frill."

EXPLANATION-GUIDED

"The image shows a close-up of a dinosaur skull, specifically that of a Triceratops."

READING

The student identifies the Triceratops, but omits the nearby person and much of the scene.

QUALITATIVE ANALYSIS

Guidance avoids a visual-path failure

GLOBAL MSE

“I'm sorry, but I cannot see any image attached to your message. Could you please provide more information or rephrase your question?”

EXPLANATION-GUIDED

“The image depicts a dinosaur skeleton on display in what appears to be a museum or exhibition hall.”

READING

Guidance restores a usable visual answer where global MSE fails to acknowledge the image.

SCOPE

Limitations

Single VLM, single dataset: one teacher, Mini-ImageNet only.

VLM judges, not humans: model-specific biases may exist.

Saliency is probe-mediated: the probe's class-discriminative focus may not entirely align with the backbone.

Teacher still dominates: the method improves fidelity but does not yet solve faithful visual-module replacement.

CONCLUSION

Explanations can be a useful distillation signal

Grad-CAM-weighted token consistently outperforms plain global MSE.

The strongest result comes from **explanation-only supervision**.

Gains manifest as **preserved local evidence** and reduced generation robustness failures.

Future work: richer prompts, human evaluation, stronger adapters, and **saliency aligned with multimodal representations**.

Thank you

Questions?